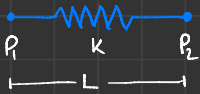


SPRINGS & DAMPER

A **spring** is an element that exerts a restoring force on masses in a dynamic system.



$$F_k = K(\bar{v}_{PR} - L)$$

$$\theta_k = \tan^{-1} \left(\frac{\bar{v}_{PR} \cdot \hat{j}}{\bar{v}_{PR} \cdot \hat{i}} \right)$$

A **damper** is an element that exerts a resistive force on masses in a dynamic system.



$$F_c = C \dot{\bar{v}}_{Q_1 Q_2} \cdot \hat{e}_{Q_1 Q_2}$$

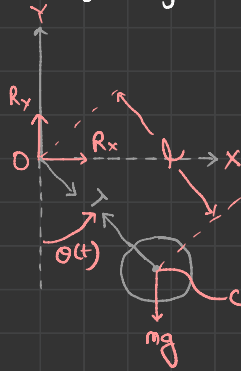
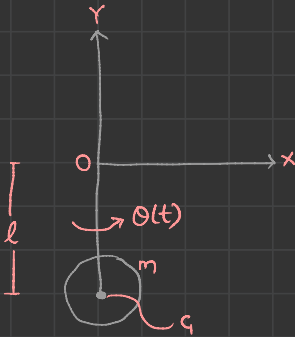
$$\theta_c(t) = \tan^{-1} \left(\frac{\bar{v}_{Q_1 Q_2} \cdot \hat{j}}{\bar{v}_{Q_1 Q_2} \cdot \hat{i}} \right)$$

$$\hat{e}_{Q_1 Q_2} = \frac{\bar{v}_{Q_1 Q_2}}{\dot{\bar{v}}_{Q_1 Q_2}}$$

Q. Derive the EOM for a simple pendulum.

A.

Free body diagram :



$$\vec{r}_{Oq} = l \sin \theta \hat{r} - l \cos \theta \hat{j}$$

$$\dot{\vec{r}}_{Oq} = l \cos \theta \dot{\theta} \hat{r} + l \sin \theta \dot{\theta} \hat{j}$$

$$\ddot{\vec{r}}_{Oq} = (l \cos \theta \ddot{\theta} - l \sin \theta \dot{\theta}^2) \hat{r} + (l \sin \theta \ddot{\theta} + l \cos \theta \dot{\theta}^2) \hat{j}$$

$$a_{ax} = l \cos \theta \ddot{\theta} - l \sin \theta \dot{\theta}^2$$

$$a_{ay} = l \sin \theta \ddot{\theta} + l \cos \theta \dot{\theta}^2$$

$$\Sigma F_{ax} = m a_{ax}$$

$$-T \sin \theta = m a_{ax}$$

$$T = \frac{-m a_{ax}}{\sin \theta}$$

$$\Sigma F_{ay} = m a_{ay}$$

$$T \cos \theta - mg = m a_{ay}$$

$$T = \frac{m a_{ay} + mg}{\cos \theta}$$

$$\therefore \frac{-m a_{ax}}{\sin \theta} = \frac{m a_{ay} + mg}{\cos \theta}$$

$$(l \sin \theta \ddot{\theta} - l \cos \theta \dot{\theta}^2) \cos \theta = \sin \theta (l \sin \theta \ddot{\theta} + l \cos \theta \dot{\theta}^2 + g)$$

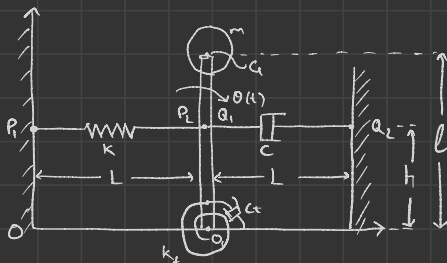
$$\cancel{\sin \theta \cos \theta \ddot{\theta}} - \cos^2 \theta \ddot{\theta} = \cancel{\sin^2 \theta \ddot{\theta}} + \cancel{\sin \theta \cos \theta \dot{\theta}^2} + \frac{g \sin \theta}{\cancel{l}}$$

$$\ddot{\Theta}(\sin^2\Theta + \cos^2\Theta) + \frac{g}{\ell} \sin\Theta = 0$$

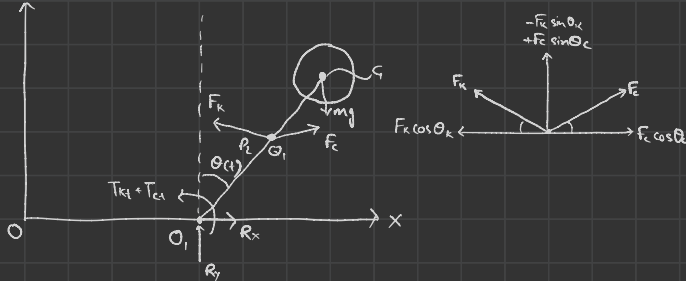
$$\ddot{\Theta} + \frac{g}{\ell} \sin\Theta = 0$$

* INVERTED PENDULUM

Q. Derive the EOM for the DOF $\theta(t)$.



A. FBD :-



$$T_{kr} = k_s \theta \quad T_{cr} = c_t \dot{\theta}$$

Forces :-

$$F_k = (\bar{r}_{P_1 P_2} - L) k$$

$$\bar{r}_{P_1 P_2} = |\bar{r}_{P_1 P_2}| = \sqrt{(\bar{r}_{P_1 P_2} \cdot \hat{i})^2 + (\bar{r}_{P_1 P_2} \cdot \hat{j})^2}$$

$$\bar{r}_{P_1 P_2} = \bar{r}_{O P_2} - \bar{r}_{O P_1}$$

$$\bar{r}_{O P_2} = (L + h \sin \theta) \hat{i} + (h \cos \theta) \hat{j}$$

$$\bar{r}_{O P_1} = h \hat{j}$$

$$\bar{r}_{P_1 P_2} = (L + h \sin \theta) \hat{i} + (h \cos \theta - h) \hat{j}$$

$$\bar{r}_{P_1 P_2} = [(L + h \sin \theta)^2 \hat{i} + (h \cos \theta - h)^2 \hat{j}]^{1/2}$$

$$F_k = \left[\left[(L + h \sin \theta)^2 \hat{i} + (h \cos \theta - h)^2 \hat{j} \right]^{1/2} - L \right] k$$

$$F_c = C(\dot{\vec{r}}_{Q_1 Q_2} \cdot \hat{e}_{Q_1 Q_2})$$

$$\vec{r}_{Q_1 Q_2} = \vec{r}_{OQ_2} - \vec{r}_{OQ_1} \quad \hat{e}_{Q_1 Q_2} = \frac{\vec{r}_{Q_1 Q_2}}{r_{Q_1 Q_2}}$$

$$r_{Q_1 Q_2} = \sqrt{(\vec{r}_{OQ_2} \cdot \hat{r})^2 + (\vec{r}_{OQ_2} \cdot \hat{j})^2}$$

$$\vec{r}_{OQ_2} = (2L\hat{r} + h\hat{j}) - ((L + \sin\theta)\hat{r} + h\cos\theta\hat{j})$$

$$\vec{r}_{Q_1 Q_2} = (L + \sin\theta)\hat{r} + (h - h\cos\theta)\hat{j}$$

$$r_{Q_1 Q_2} = \left((L + \sin\theta)^2 \hat{r} + (h - h\cos\theta)^2 \hat{j} \right)^{1/2}$$

$$\hat{e}_{Q_1 Q_2} = \frac{(L + \sin\theta)\hat{r} + (h - h\cos\theta)\hat{j}}{\left((L + \sin\theta)^2 \hat{r} + (h - h\cos\theta)^2 \hat{j} \right)^{1/2}}$$

$$\dot{\vec{r}}_{Q_1 Q_2} = \dot{\theta}\cos\theta\hat{r} + h\dot{\theta}\sin\theta\hat{j}$$

$$F_c = C \left[(\dot{\theta}\cos\theta\hat{r} + h\dot{\theta}\sin\theta\hat{j}) \cdot \frac{(L + \sin\theta)\hat{r} + (h - h\cos\theta)\hat{j}}{\left((L + \sin\theta)^2 \hat{r} + (h - h\cos\theta)^2 \hat{j} \right)^{1/2}} \right]$$

Newton's law :-

$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$

$$\vec{r}_{OQ_1} = (L + h\sin\theta)\hat{r} + h\cos\theta\hat{j}$$

$$\dot{\vec{r}}_{OQ_1} = h\dot{\theta}\cos\theta\hat{r} - h\dot{\theta}\sin\theta\hat{j}$$

$$\ddot{\vec{r}}_{OQ_1} = (h\ddot{\theta}\cos\theta - h\dot{\theta}^2\sin\theta)\hat{r} - (h\ddot{\theta}\sin\theta + h\dot{\theta}^2\cos\theta)\hat{j}$$

$$\ddot{\vec{r}}_{OQ_1} = (h\ddot{\theta}\cos\theta - h\dot{\theta}^2\sin\theta)\hat{r} + (-h\ddot{\theta}\sin\theta - h\dot{\theta}^2\cos\theta)\hat{j}$$

↓
 a_{ax}

↓
 a_{ay}

$$\Sigma F_x = m a_{gx}$$

$$-F_K \cos \theta_K + F_C \cos \theta_C + R_x = m (h \ddot{\theta} \cos \theta - h \dot{\theta}^2 \sin \theta)$$

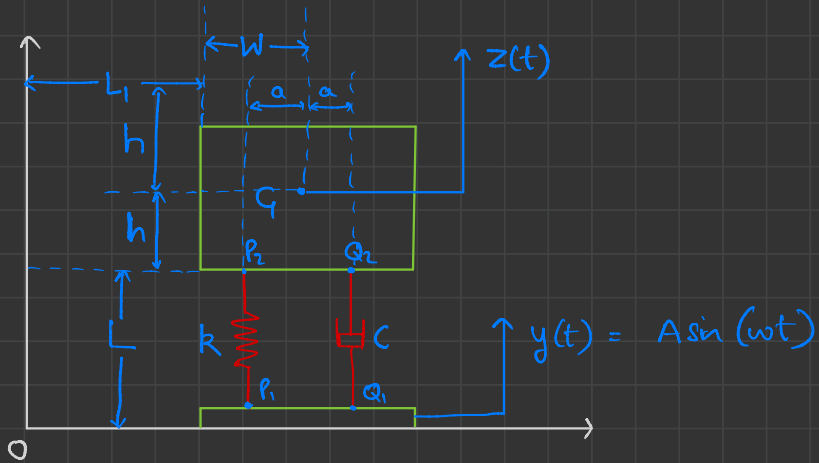
$$\Sigma F_y = m a_{gy}$$

$$-F_K \sin \theta_K + F_C \sin \theta_C + R_y - mg = m (-h \ddot{\theta} \sin \theta - h \dot{\theta}^2 \cos \theta)$$

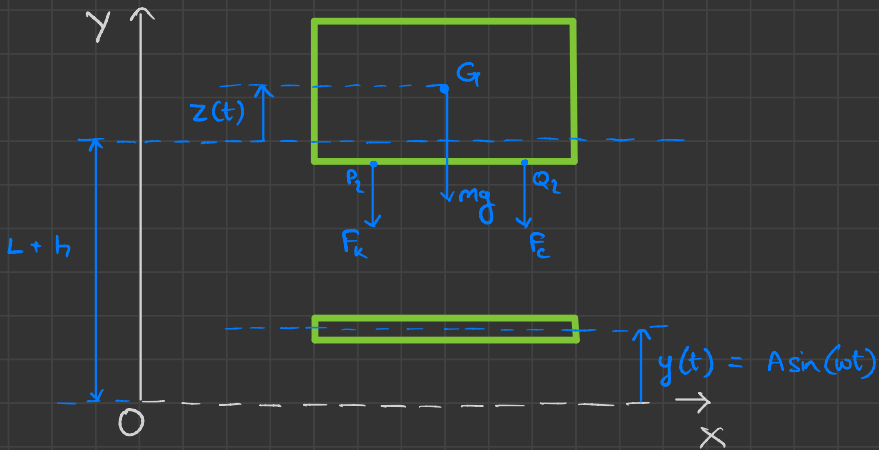
$$\Sigma M_{O_1} = I_{O_1}(\ddot{\theta})$$

$$\begin{aligned} \Sigma M_{O_1} &= -m l^2 \ddot{\theta} \\ &= (-F_K \sin \theta_K + F_C \sin \theta_C) h \sin \theta + F_K \cos \theta_K h \cos \theta \\ &\quad - F_C \cos \theta_C h \cos \theta - mg l \sin \theta + K l \theta + G l \dot{\theta} \end{aligned}$$

Base Excitation



FBD :-



$$\vec{r}_{O/G} = (L+h)\hat{j} + (L+w)\hat{i} + z(t)\hat{j}$$

$$\dot{\vec{r}}_{O/G} = \dot{z}\hat{j} + 0\hat{i}$$

$$\ddot{\vec{r}}_{O/G} = \ddot{z}\hat{j} + 0\hat{i} \quad (\text{Acceleration})$$

Forces

$$F_k = k (\bar{r}_{P_1 P_2} - L)$$

$$\bar{r}_{P_1 P_2} = \bar{r}_{O P_2} - \bar{r}_{O P_1}$$

$$\bar{r}_{O P_1} = (L_1 + w - a)\hat{i} + y(t)\hat{j}$$

$$\bar{r}_{O P_2} = (L_1 + w - a)\hat{i} + (L + z(t))\hat{j}$$

$$\bar{r}_{P_1 P_2} = (z(t) - y(t) + L)\hat{j}$$

$$\bar{r}_{P_1 P_2} = z(t) - y(t) + L$$

So, $F_k = k (z(t) - y(t))$

$$F_c = c (\dot{z}(t) - \dot{y}(t))$$

$$= c \dot{\bar{r}}_{Q_1 Q_2} \cdot \hat{e}_{Q_1 Q_2}$$

$$\hat{e}_{Q_1 Q_2} = \hat{j} = \frac{\bar{r}_{Q_1 Q_2}}{r_{Q_1 Q_2}}$$

Newton's Laws

$$\sum F_y = m a_y$$

$$-mg - k(z - y) - c(\dot{z} - \dot{y}) = m \ddot{z}$$

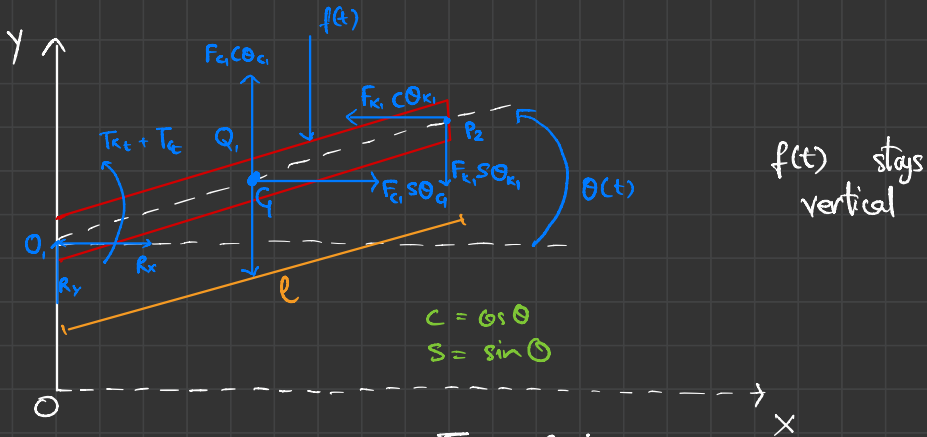
$$m \ddot{z} + c \dot{z} + k z = -mg + c \dot{y} + k y$$

$$z(t) = z_1(t) + \Delta_{st}$$

$$m \ddot{z}_1 + c \dot{z}_1 + k (\cancel{\Delta_{st}} + z_1) = \cancel{-mg} + c \dot{y} + k y$$

HORIZONTAL BAR

... continuation



$$T_{kt} = k_t \theta$$

$$T_{ct} = c_t \dot{\theta}$$

$\theta(t)$ is small

$$F_{k1} = k_1 l \theta$$

$$F_{c1} = -c_1 \frac{l}{2} \dot{\theta}$$

Acceleration

$$\vec{r}_{OQ_1} = L \hat{j} + \frac{l}{2} \cos \theta \hat{i} + \frac{l}{2} \sin \theta \hat{j}$$

$$\dot{\vec{r}}_{OQ_1} = -\frac{l}{2} \dot{\theta} (\sin \theta) \hat{i} + \frac{l}{2} \dot{\theta} (\cos \theta) \hat{j}$$

$$\begin{aligned} \ddot{\vec{r}}_{OQ_1} = & \left(-\frac{l}{2} \ddot{\theta} \sin \theta - \frac{l}{2} \dot{\theta}^2 \sin \theta \right) \hat{i} \\ & + \left(\frac{l}{2} \ddot{\theta} \cos \theta - \frac{l}{2} \dot{\theta}^2 \sin \theta \right) \hat{j} \end{aligned}$$

$$a_{Q_1 x} = \ddot{\vec{r}}_{OQ_1} \cdot \hat{i}$$

$$a_{Q_1 y} = \ddot{\vec{r}}_{OQ_1} \cdot \hat{j}$$

Newton's Laws

$$\Sigma F_x = ma_{ax}$$

$$\Sigma M_a = + I_a \ddot{\theta}$$

$$I_a = \frac{1}{12} m l^2$$

$$\Sigma F_y = ma_{ay}$$

$$\Sigma M_{O_1} = + I_{O_1} \ddot{\theta}$$

$$I_{O_1} = \frac{1}{3} m l^2$$

$$\Sigma F_x :-$$

$$R_x + F_{c1} \cos \theta_{c1} - F_{k1} \cos \theta_{k1} = -m \left(\frac{l}{2} \ddot{\theta} \sin \theta + \frac{l}{2} \dot{\theta}^2 \sin \theta \right)$$

$$\Sigma F_y :-$$

$$+R_y - mg + F_{c1} \sin \theta_{c1} - f(t) - F_{k1} \sin \theta_{k1} = m \left(\frac{l}{2} \ddot{\theta} \cos \theta - \frac{l}{2} \dot{\theta}^2 \sin \theta \right)$$

$$\Sigma M_a :-$$

$$\Sigma M_a = I_a \ddot{\theta}$$

$$\frac{1}{12} m l^2 \ddot{\theta} = -T_{kt} - T_{ct} - R_y \frac{l}{2} \cos \theta + R_x \frac{l}{2} \sin \theta - F_{k1} \sin \theta_{k1} \frac{l}{2} \cos \theta + F_{k1} \cos \theta_{k1} \frac{l}{2} \sin \theta - f(t) a \cos \theta$$

$$\Sigma M_{O_1} :-$$

$$\Sigma M_{O_1} = I_{O_1} \ddot{\theta}$$

$$\frac{1}{3} m l^2 \ddot{\theta} = -T_{kt} - T_{ct} - mg \frac{l}{2} \cos \theta + F_{c1} \sin \theta_{c1} \frac{l}{2} \cos \theta - F_{c1} \cos \theta_{c1} \frac{l}{2} \sin \theta - f(t) \left[\frac{l}{2} + a \right] \cos \theta$$

$$- F_{K1} \sin \theta_{K1} l \cos \theta + F_{K1} \cos \theta_{K1} l \sin \theta$$

when θ is small , $\cos \theta = 1$

$$\sin \theta = 0$$

$$\sin \theta_{K1} = 1$$

$$\cos \theta_{K1} = 0$$

$$\sin \theta_c = 1$$

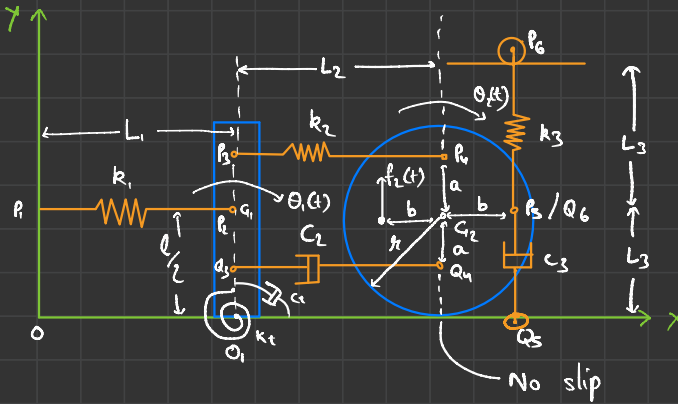
$$\cos \theta_c = 0$$

Equations of Motion :

$$\frac{1}{3} m l^2 \ddot{\theta} = -T_{K1} - T_{c1} - mg \frac{l}{2} + F_{c1} \frac{l}{2} - f(t) \left(\frac{l}{2} + a \right) - F_{K1} l$$

Differential Equation governing the motion of the bar

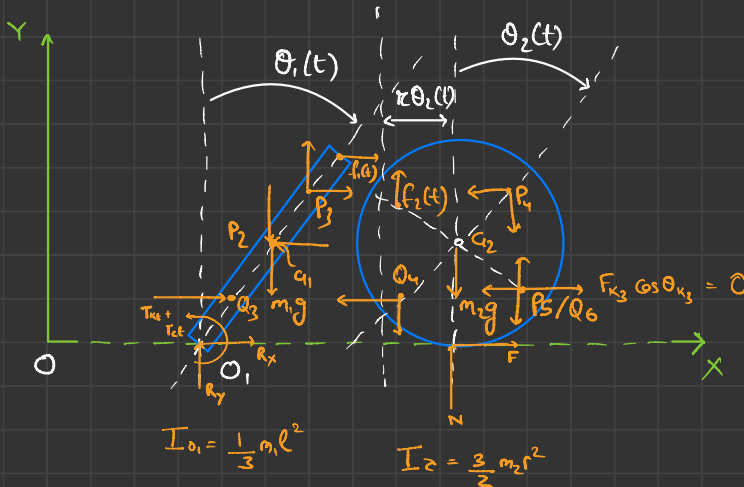
INDUCED VIBRATIONS



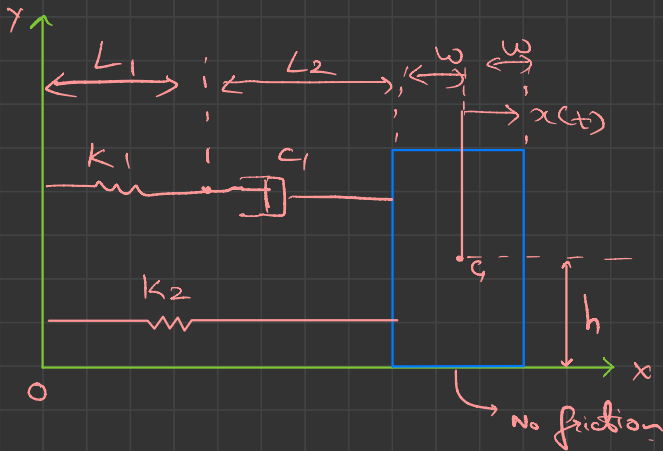
$f_1(t)$ stays horizontal
 $f_2(t)$ stays vertical

$$\theta_i(t) = A \cos(\omega t)$$

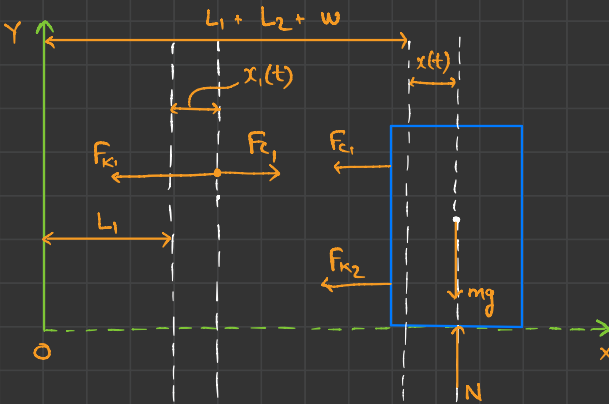
FREE BODY DIAGRAM



1.5 DoF



FREE BODY DIAGRAM



$$F_{k1} = k_1 x_1$$

$$F_{k2} = k_2 x_2$$

$$F_{c1} = c_1 (\dot{x} - \dot{x}_1)$$

$$c_1 (\dot{x} - \dot{x}_1) = k_1 x_1$$

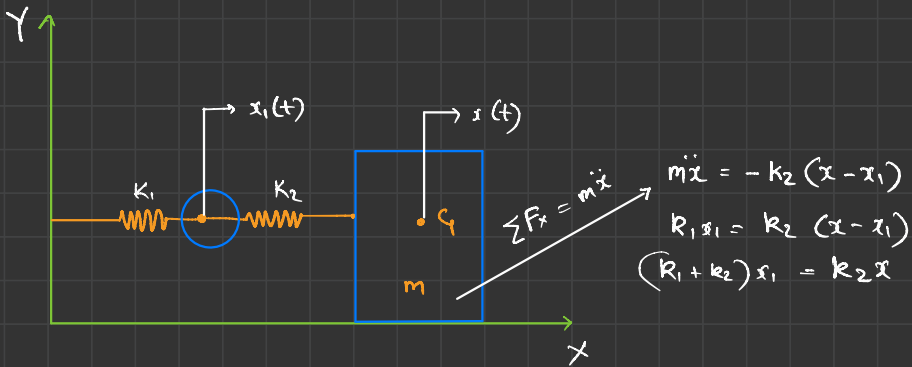
For mass m ,

$$\sum F_x = m\ddot{x}$$

$$m\ddot{x} = -F_{k_2} - F_{c_1}$$

$$m\ddot{x} = -k_2 x - c_1 (\dot{x} - \dot{x}_1)$$

$$m\ddot{x} + c_1 \dot{x} + k_2 x - c_1 \dot{x}_1 = 0$$



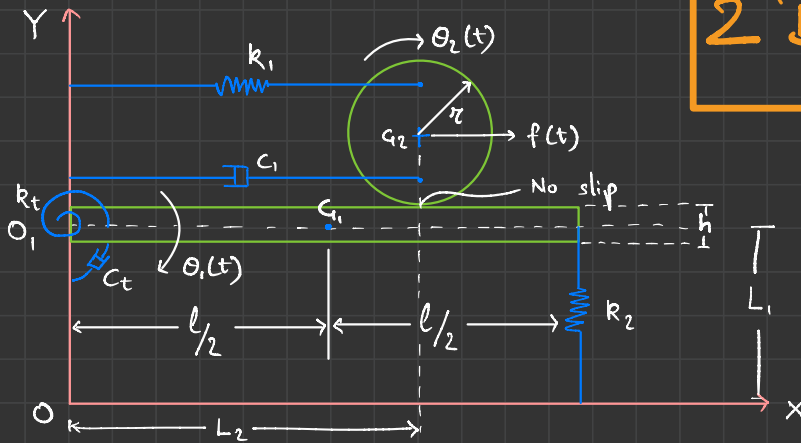
$$x_1 = \left(\frac{k_2}{k_1 + k_2} \right) x$$

$$m\ddot{x} + k_2 \left(x - \frac{k_2}{k_1 + k_2} x \right) = 0$$

$$m\ddot{x} + \left(\frac{k_1 k_2}{k_1 + k_2} \right) x = 0$$

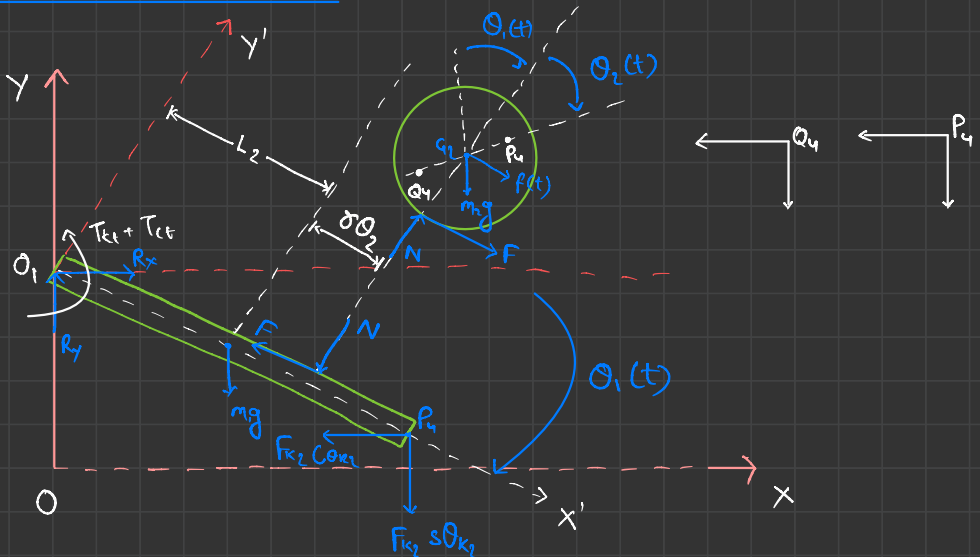
$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

2 DoF



$f(t)$ stays parallel to the bar

FREE BODY DIAGRAM :-



Rotation of Cylinder : $-(\theta_1(t) + \theta_2(t)) \hat{k}$

$$\begin{aligned} \hat{i}' &= \cos \theta_1 \hat{i} - \sin \theta_1 \hat{j} \\ \hat{j}' &= \sin \theta_1 \hat{i} + \cos \theta_1 \hat{j} \end{aligned}$$

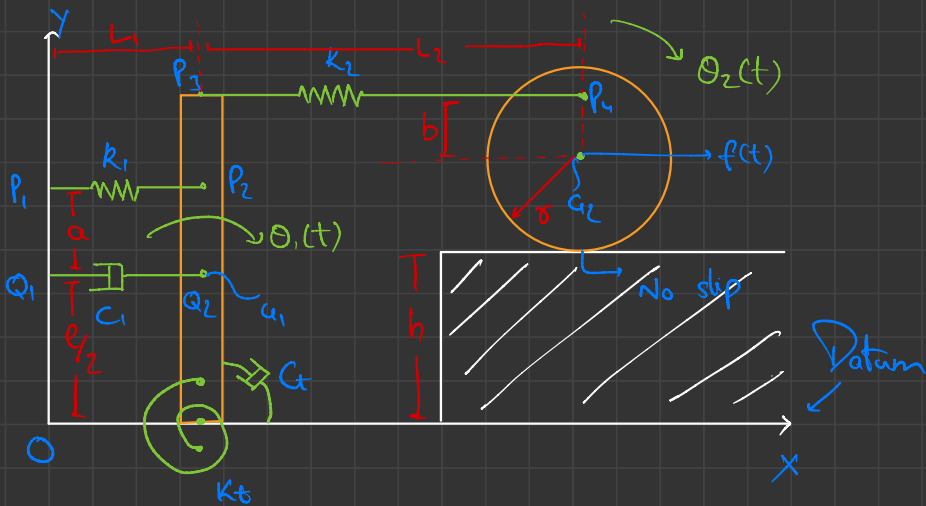
$$\begin{aligned} \dot{\hat{i}}' &= -\dot{\theta}_1 \hat{j} \\ \dot{\hat{j}}' &= \dot{\theta}_1 \hat{i} \end{aligned}$$

$$\bar{\sigma}_{op_1} = (L_2 + \sigma\theta_2)\hat{r}' + L_1\hat{j} + \frac{h}{2}\hat{j} + \sigma\hat{j}' + a\cos\theta_2\hat{j}' + a\sin\theta_2\hat{i}'$$

POWER & ENERGY

Kinetic Energy, Potential Energy, Power dissipated

$$T = \frac{1}{2} m_i \overline{\dot{\theta}_{O_{i1}}} \cdot \overline{\dot{\theta}_{O_{i1}}}$$

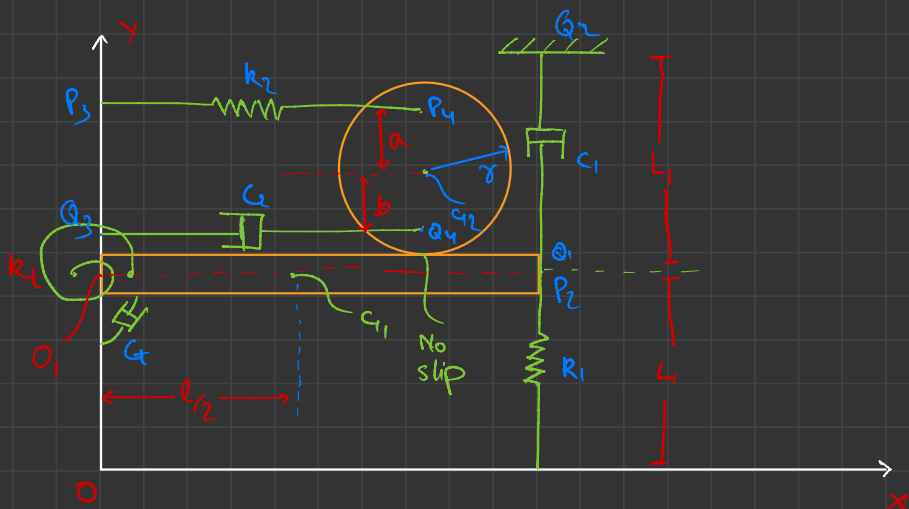


$$KE = \underbrace{\frac{1}{2} m_1 \overline{\dot{\theta}_{O_{11}}} \cdot \overline{\dot{\theta}_{O_{11}}} + \frac{1}{2} I_{O_1} \dot{\theta}_1^2}_{\frac{1}{2} I_{O_1} \dot{\theta}_1^2} + \frac{1}{2} m_2 \overline{\dot{\theta}_{O_{21}}} \cdot \overline{\dot{\theta}_{O_{21}}} + \frac{1}{2} I_{O_2} \dot{\theta}_2^2$$

$$I_{O_1} = I_{O_1} + m_1 \left(\frac{h}{2}\right)^2$$

$$PE = \frac{1}{2} k_1 (\overline{\theta_{P_1 P_2}} - L_1)^2 + \frac{1}{2} k_2 (\overline{\theta_{P_3 P_4}} - L_2)^2 + \frac{1}{2} k_t \theta_1^2 + m_1 g \overline{\theta_{O_{11}}} \cdot \hat{j} + m_2 g \overline{\theta_{O_{21}}} \cdot \hat{j}$$

$$P_{diss} = c_1 (\dot{\delta}_{\theta_1, \theta_2} \cdot \hat{e}_{\theta_1, \theta_2})^2 + c_t \dot{\theta}_1^2$$



$$(KE)_{cyl} = \frac{1}{2} m_2 \dot{\delta}_{\theta_1, \theta_2} \cdot \dot{\delta}_{\theta_1, \theta_2} + \frac{1}{2} I_{Q_2} (\dot{\theta}_1 + \dot{\theta}_2)^2$$

$$(KE)_{BAR} = \frac{1}{2} m_1 \dot{\delta}_{\theta_1} \cdot \dot{\delta}_{\theta_1} + \frac{1}{2} I_{O_1} \dot{\theta}_1^2 = \frac{1}{2} I_{O_1} \dot{\theta}_1^2$$

TEE

$$\hat{z}' = \cos \theta_1 \hat{i} - \sin \theta_1 \hat{j}$$

$$\hat{y}' = \sin \theta_1 \hat{i} + \cos \theta_1 \hat{j}$$

$$\hat{z}' = \cos \theta_1 \hat{i} - \sin \theta_1 \hat{j}$$

$$\hat{y}' = \sin \theta_1 \hat{i} + \cos \theta_1 \hat{j}$$

$$t_{kt} = k_t \theta$$

$$t_{ct} = c_t \dot{\theta}$$

$$F_{c1} = k(\bar{r}_{AP1} - L_1)$$

$$\bar{\sigma}_{OP1}, \bar{\sigma}_{OP2}, \bar{\sigma}_{AP1} = \bar{\sigma}_{OP1} - \bar{\sigma}_{OP2}$$

$$\bar{\sigma}_{AP2} = \left[(\bar{\sigma}_{AP1} \cdot \hat{i})^2 + (\bar{\sigma}_{AP2} \cdot \hat{j})^2 \right]^{1/2}$$

$$\theta_{K1}(t) = \tan^{-1} \left(\frac{\bar{\sigma}_{P1P2} \cdot \hat{j}}{\bar{\sigma}_{P1P2} \cdot \hat{i}} \right)$$

$$F_{c1} = c_1 \dot{\bar{\sigma}}_{Q1Q2} \cdot \hat{e}_{Q1Q2}$$

$$\hat{e}_{Q1Q2} = \frac{\bar{\sigma}_{Q1Q2}}{\sigma_{Q1Q2}}$$

$$KE = \frac{1}{2} m_1 \bar{\sigma}_{O41} \cdot \bar{\sigma}_{O41} + \frac{1}{2} m_2 \bar{\sigma}_{O42} \cdot \bar{\sigma}_{O42} + \frac{1}{2} I_{O1} \dot{\theta}_1^2 + \frac{1}{2} I_{O2} \dot{\theta}_2^2$$

$$PE = \frac{1}{2} k_1 (\bar{\sigma}_{AP1} - L_1)^2 + \frac{1}{2} k_2 (\bar{\sigma}_{AP2} - L_2)^2 + \frac{1}{2} k_t \theta_1^2 + m_1 g \bar{\sigma}_{O41} \cdot \hat{j} + m_2 g \bar{\sigma}_{O42} \cdot \hat{j}$$

$$P_{in} = f_t(t) \hat{i} \cdot \dot{\bar{\sigma}}_{O42} + M(t) \hat{k} \cdot \dot{\theta}_1 \hat{k}$$

$$P_{diss} = (c_t \dot{\theta}_1) \dot{\theta}_1 + c_1 (\dot{\bar{\sigma}}_{Q1Q2} \cdot \hat{e}_{Q1Q2})^2 + c_2 (\dot{\bar{\sigma}}_{Q3Q4} \cdot \hat{e}_{Q3Q4})^2$$

